

Dr Alhafidz Hamdan

ECAT SCREDS Clinical Lecturer and Honorary Resident in Histopathology
University of Edinburgh & NHS Lothian
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University Education

- PhD (Cancer Genetics), University of Edinburgh (2025)
- MBBS, Newcastle University (2013)
- Intercalated MRes (Neuroscience), Newcastle University (2011)

Career since Graduation

- **ECAT SCREDS Clinical Lecturer and Honorary Resident (Histopathology)**, University of Edinburgh & NHS Lothian, 2026–present
- **Specialist Training Programme (Histopathology)**, NHS Education for Scotland, 2024–present
- **ECAT CRUK Clinical Research Fellow**, University of Edinburgh, 2019–2022
- **Specialist Training Programme (Neurosurgery)**, NHS Education for Scotland, 2018–2024
- **Junior Clinical Fellow (Neurosurgery)**, King's College Hospital, London, 2017–2018
- **Core Surgical Training Programme**, Health Education Yorkshire and the Humber, 2015–2017
- **Academic Foundation Programme (Neurosurgery)**, Health Education North East, 2013–2015

Background

Alhafidz completed a CRUK-linked ECAT PhD in Cancer Genetics with Professors Steven Pollard and Colin Semple. With Pollard he developed synthetic super-enhancers for viral gene therapy in glioblastoma, now in a Phase 0/1 trial. With Semple, his research focused on genomic instability — aneuploidy, complex structural rearrangements and extrachromosomal DNA (ecDNA) in glioblastoma. From more than 15,000 cancer whole genomes in the 100,000 Genomes Project, he built EpiTracer to detect episomal ecDNA and study its origins. As a student and now postdoctoral clinical research associate, he has contributed bioinformatics to projects in high-grade serous ovarian cancer and mesothelioma. As an aspiring academic neuropathologist, he combines computational biology, histology and genetics to study brain-tumour evolution and ecDNA formation. The ECAT SCREDS Clinical Lectureship supports this work alongside histopathology training, with the long-term aim of leading an academic neuropathology laboratory as an honorary NHS consultant.

Key Recent Publications

h-index 26 · i10-index 39 · 53 publications · 5,120 citations (16 August 2026)

- Koeber U, Matjusaitis M, White R, Alfazema N, Furlong K, Wang Z, Hamdan A, Morisse G, Navarette M, Wang J, Clark MP, Blin C, Dewari P, Hong HI, Sheraz S, Dewari P, Southgate B, Mendrychowski J, Kho V, Gogolok S, Morrison GM, Cancino FG, Robertson F, Williams A, Rosser S, Brennan P, Sieger D, Soufi A, Quezada S, Pollard S. Synthetic super-enhancers enable precision viral immunotherapy. *Nature*. 2026 May;653(8113):232-241. PMID 41951744
- Hamdan A, Ewing A. Unravelling the tumour genome: The evolutionary and clinical impacts of structural variants in tumorigenesis. *J Pathol*. 2022 Jul;257(4):479-493. PMID 35355264
- Ewing A, Meynert A, Silk R, Aitken S, Bendixsen D, Churchman M, Brown S, Hamdan A, Mattocks J, Grimes GR, Ballinger T, Hollis RL, Herrington CS, Thomson JP, Sherwood K, Parry T, Eisri-Bloom E, Bartos C, Croy I, Ferguson M, Lennie M, McGoldrick T, McPahil N, Siddiqui N, Glasspool R, Mackean M, Nussey F, McDade B, Ennis D, The Scottish Genomes Partnership, McMahon L, Matakidou A, Dougherty B, March R, Barrett JC, McNeish IA, Biankin AV, Roxburgh P, Gourley C, Semple CA. Divergent trajectories to structural diversity impact patient

survival in high grade serous ovarian cancer. *Nat Commun.* 2025 Jul 1;16(1):5586. PMID 40593568

- Williams CAC, Djeghloul DZ, Veland N, Hamdan A, Kalantzaki M, Lo E, Illingworth R, Kriegsheim A, Fisher AG, Soufi A, Pollard S. SOX2 phosphorylation during mitosis limits genomic damage. *Genes Dev.* 2026 Feb 2;40(3-4):185-198. PMID 41213799
- Purshouse K, Fridman E, Boyle S, Dewari P, Grant V, Hamdan A, Morrison G, Brennan P, Beentjes S, Pollard S, Bickmore W. Oncogene expression from extrachromosomal DNA is driven by copy number amplification and does not require spatial clustering in glioblastoma stem cells. *eLife.* 2022 Dec 7;11:e80207. PMID 36476408

Awards & Distinctions

- Editorial reviewer, Nature Communications (2026–present)
- Second prize, oral presentation, Royal Society of Medicine Pathology Society Conference (2025)
- British Division of the International Academy of Pathology (BDIAP) Award to attend a molecular pathology course (2025)
- JMJ Travel Award to present PhD work at the EMBO Chromoploidy Conference, Italy (2025)
- Editorial reviewer, British Journal of Neurosurgery (2020–2023)
- ECAT CRUK Clinical Research Fellow, University of Edinburgh (2019–2022)
- MBBS (Honours), Newcastle University (2013)
- Rupert Kelly Prize in Psychiatry, Newcastle University (2012)
- Luke Armstrong Prize in Experimental Pathology, Newcastle University (2012)
- Distinctions in Intercalated MRes (Neuroscience), Newcastle University (2010)

Leadership & Management

- Histopathology resident representative, Clinical Governance committee (2024–present)
- Local lead coordinator for national and international surgical research studies, including CRANIAL (CSF Rhinorrhoea After Endonasal Intervention to the Skull Base), GNOS (Global Neurotrauma Outcomes Study) and ISOS (International Surgical Outcome Study) (2014–2021)
- President, National Academic Medicine Student Conference (2012)
- President, Newcastle Widening Participation Programme (2012)

Teaching Commitments

- Trainee lead, Undergraduate Medical Education Curriculum in Histopathology (2026–present)
- Dissertation examiner (Bachelor's/Master's), University of Edinburgh (2021–present)
- OSCE examiner, Newcastle University (2013–2017)
- Neuroanatomy demonstrator, Newcastle University (2014–2015)
- Clinical Skills Tutor, Newcastle University (2013–2014)